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$$I. a_{th} = 10 \times 10^{-6}$$

$$\dot{F} = 8 \times 10^{13} \text{ fissions/cm}^3\text{-s}$$

$$T_f = 1,500 \text{ K}$$

$$T_{ref} = 300 \text{ K}$$

$$\Delta p_0 = 0.015$$

$$B_0 = 5 \text{ MWth/kgU}$$

$$\rho(\text{CO}_2) = 10.97 \text{ g/cc}$$

$$t = 300 \text{ d}$$

$$N_A = 6.022 \times 10^{23} \text{ atoms/mol}$$

$$M_{\text{CO}_2} = 270 \text{ g/mol}$$

$$N_V = \frac{\rho(\text{CO}_2) N_A}{M_{\text{CO}_2}}$$

$$N_V = 2.446716296 \times 10^{22}$$

$$\beta = \frac{\dot{F} \times t}{N_V}$$

$$\beta = 0.0847503245$$

$$B_0 = 0.005263579$$

$$C_D = 1$$

$$\epsilon_D = \Delta p_0 \left(e^{\frac{B_0 \Delta p_0}{C_D B_0}} - 1 \right)$$

$$\epsilon_D = 0.015$$

$$\epsilon_{SF} = 5.577 \times 10^{-2} \beta$$

$$\epsilon_{SF} = 0.0518449858$$

$$\epsilon_{GF} = 1.96 \times 10^{-28} \beta (2,800 - T)^{11.73} \times e^{-0.0162(2,800 - T)} e^{-17.8 \beta}$$

$$\epsilon_{GF} = 2.34341719 \times 10^{-8}$$

$$\epsilon_t = \epsilon_D + \epsilon_{SF} + \epsilon_{GF} + \epsilon_{TH}$$

$$\epsilon_{TH} = 0.07$$

$$\boxed{\epsilon_t = 0.03685007}$$

$$2. \sigma = 60 \text{ MPa}$$

$$T = 600 \text{ K}$$

$$\text{LHR} = 250 \text{ W/cm}$$

$$t = 200 \text{ d}$$

$$A_0 = 3.14 \times 10^{24}$$

$$G = 4.2519 \times 10^{10} - 2.2185 \times 10^{-7} T$$

$$n = 5$$

$$Q = 2.7 \times 10^5 \text{ J/mol}$$

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$$C_0 = 1.654 \times 10^{-24}$$

$$C_1 = 0.85$$

$$C_2 = 1$$

$$G = 2.9208 \times 10^{10} \text{ Pa}$$

$$R = 8.314 \text{ J/mol-K}$$

$$\dot{E}_{th} = A_0 \frac{1}{2} n e^{-\frac{Q}{RT}}$$

$$\dot{E}_{th} = 3.5788199 \times 10^{-13} \text{ J}$$

$$\Phi = 3 \times 10^{11} \times \text{LHR}$$

$$\Phi = 7.5 \times 10^{13}$$

$$\dot{E}_{ir} = C_0 \Phi^{C_1} G^{C_2}$$

$$\dot{E}_{ir} = 6.172894143 \times 10^{-5} \text{ J}$$

→ nope ...

Wrong units?

$$\dot{E}_t = \dot{E}_{th} + \dot{E}_{ir}$$

$$\dot{E}_t = 6.172894179 \times 10^{-5} \text{ J}$$

$$E_t = \dot{E}_t \times t$$

$$E_t = 1,066.67611$$

$$3.7 = 250 \text{ d}$$

$$T = 1,400 \text{ K}$$

$$\dot{F} = 2 \times 10^{13} \text{ fissions/cm}^3\text{-s}$$

$$a_1 = 5 \times 10^{-6} \text{ m}$$

$$a_2 = 40 \times 10^{-6} \text{ m}$$

$$k_B = 1.38 \times 10^{-23} \text{ J/K}$$

$$D_1 = 7.6 \times 10^{-6} \exp\left(-\frac{3.03}{k_B T}\right)$$

$$D_1 = 9.31264184 \times 10^{-17} \text{ cm}^2/\text{s}$$

$$D_2 = 1.41 \times 10^{-18} \exp\left(-\frac{1.19}{k_B T}\right) \text{ cm}^2/\text{s}$$

$$D_2 = 3.26787168 \times 10^{-16} \text{ cm}^2/\text{s}$$

$$D_3 = 2.0 \times 10^{-30} \text{ cm}^2/\text{s}$$

$$D_3 = 4 \times 10^{-17} \text{ cm}^2/\text{s}$$

$$D = D_1 + D_2 + D_3$$

$$D = 4.55913587 \times 10^{-16} \text{ cm}^2/\text{s}$$

$$\tau_1 = \frac{a_1^2}{D}$$

$$\tau_1 = 0.1397365339$$

$$\tau_1 < \pi^{-2}$$

$$f_1 = 4 - \frac{D_1}{\pi a_1^2} - 3 \frac{D_2}{a_1^2}$$

$$f_1 = 0.32973$$

-mixed in-pile + postwired eqns.

$$\tau_2 = \frac{a_2^2}{D}$$

$$\tau_2 = 6.154833424 \times 10^{-44}$$

$$\tau_2 < \pi^{-2}$$

$$f_2 = 4 - \frac{D_1}{\pi a_2^2} - 3 \frac{D_2}{a_2^2}$$

$$f_2 = 0.154141$$

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4. The typical charge state is $+4$. The introduction of fission product with charges other than $+4$ can cause it to change as the lattice must maintain charge neutrality. $\frac{4}{4}$

5. The oxygen concentration increases as the fuel pellet radius increases. Ma impacts this as it is the limiting factor in how the oxygen concentration changes due to the Gibbs free energy of Ma and oxygen. $\frac{3}{5}$

6. 2. thermal conductivity -

2. fission gas release $\frac{3}{3}$

3. thermal expansion -

7. 1. Soluble oxides that decrease thermal conductivity and cause swelling

2. Insoluble oxides that cause swelling.

3. Metals that increase thermal conductivity.

$\frac{10}{10}$ 4. Volatile gases that decrease thermal conductivity and cause swelling

5. Noble gases that decrease thermal conductivity and cause swelling and can increase the plenum pressure if released.

8. 1. Accumulation where fission gases are created and form bubbles that move toward dislocations. - toward grain boundaries, gas diffuses

$\frac{5}{8}$ 2. Fission gas movement in dislocations and grain boundaries. - bubble growth on grains

3. Fission gas movement to free surfaces where fission gas bubbles either stay or are released.

9. Creep is slow strain when $\sigma < \sigma_y$ that can result in failure.

$\frac{5}{5}$ Dislocation creep is due to the movement and gliding of dislocations in the material lattice.

$\frac{7}{8}$ 10. High burnup structure is the more porous and smaller grain boundary structure of UO_2 fuel pellets at the periphery of the pellet due to higher power, lower temperature, and burnup. It has increased thermal conductivity and retention of more fission gases. - why form?

3/6 11. The hydride ~~form~~ and expand the Zr lattice. Irradiation growth occurs in the basal plane. The parameter a increases and the parameter c decreases. - point defects!

1/8 12. Creep causes the Zr cladding to become more ductile and less hard. - not what I wanted...

7/12 13. 1. Corrosive environment - fission products
2. susceptible material - Zr
3. sufficient stress - pressure
4. sufficient time - operation time
- needed more on all of these

3/5 14. It causes corrosion due to CrI and ZrI₂
- needed more...

